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**RĪGAS ZIEMEĻVALSTU UNIVERSITĀTE
(RNU)**

**Informācijas sistēmu integrācijas brieduma novērtējums
projektu vadības birojos: pierādījumi no Latvijas uzņēmumiem**

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**Assessment of Information System Integration Maturity in
Project Management Offices: Evidence from Latvian
Enterprises**

BACHELOR'S THESIS

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Anotācija

Šis bakalaura darbs pēta informācijas sistēmu integrācijas briedumu projektu vadības biroju (PVB) vidē Latvijas vidējos un lielos uzņēmumos. Tā kā organizācijas arvien vairāk paļaujas uz vairākiem digitālajiem rīkiem projektu plānošanai, pārskatošanai un pārvaldībai, sistēmu integrācijas pakāpei ir tieša ietekme uz datu kvalitāti, pārskatu efektivitāti un pārvaldības standartizāciju. Neskatoties uz tās praktisko nozīmi, šī tēma Latvijas kontekstā ir saņēmusi ierobežotu empīrisku uzmanību.

Bakalaura darbs sastāv no četrām nodaļām. 1. nodaļā aplūkoti teorētiskie pamati, definējot galvenos jēdzienus, piemēram, PVB, informācijas sistēmu integrāciju, brieduma modeļus, biznesa inteliģenci un datu kvalitāti, kā arī apskatīti esošie pētījumi par PVB digitalizāciju un integrācijas brieduma ietvariem. 2. nodaļā sniegta pētījuma metodoloģija, iekļaujot četru līmeņu Integrācijas Brieduma Modeļa izstrādi un strukturēta aptaujas instrumenta veidošanu. 3. nodaļā veikts empīriskais pētījums, prezentējot aptaujas rezultātus, kas iegūti no Latvijas uzņēmumu pārstāvjiem ar formālām PVB struktūrām, papildināts ar ekspertu intervijām un statistisko analīzi. 4. nodaļā izstrādāti praktiski ieteikumi informācijas sistēmu integrācijas brieduma uzlabošanai PVB vidē, balstoties uz empīriskajiem rezultātiem.

Pētījuma metodoloģiskais pamats ietver sistemātisku literatūras apskatu, aptaujas bāzētu kvantitatīvo pētījumu, strukturētas ekspertu intervijas, aprakstošo statistisko analīzi, kā arī korelācijas un regresijas analīzi.

Paredzams, ka pētījums atklās, ka lielākā daļa PVB vižu Latvijā darbojas vidējā integrācijas brieduma līmenī, un ka augstāks integrācijas briedums ir saistīts ar lielāku pārskatu automatizāciju, labāku datu kvalitāti un stiprāku pārvaldības standartizāciju. Rezultāti ir paredzēti, lai sniegtu uz pierādījumiem balstītus norādījumus organizācijām, kas vēlas uzlabot savas projektu vadības informācijas sistēmas.

Bakalaura darbs izstrādāts **XX** lapaspusēs un ietver **YY** tabulas, **ZZ** attēlus, **xy** algoritmus, **yz** programmu kodu izdrukas (listings), **xz** pielikumus, kā arī literatūras sarakstu ar **zx** avotiem.

Atslēgvārdi: Projektu vadības birojs, Informācijas sistēmu integrācija, Integrācijas brieduma modelis, Pārskatu automatizācija, Datu kvalitāte, Biznesa inteliģence, Pārvaldības standartizācija.

Abstract

This Bachelor's thesis examines the integration maturity of information systems within Project Management Office (PMO) environments in medium and large Latvian enterprises. As organizations increasingly rely on multiple digital tools for project planning, reporting, and governance, the degree to which these systems are integrated has a direct impact on data quality, reporting efficiency, and governance standardization. Despite its practical relevance, this topic has received limited empirical attention in the Latvian context.

The Bachelor Paper consists of four chapters. Chapter 1 reviews the theoretical background, defining key concepts such as PMO, information system integration, maturity models, business intelligence, and data quality, and surveys existing research on PMO digitalization and integration maturity frameworks. Chapter 2 presents the research methodology, including the design of a four-level Integration Maturity Model and the development of a structured survey instrument. Chapter 3 contains the empirical study, presenting the results of a survey conducted among representatives of Latvian enterprises with formal PMO structures, supplemented by expert interviews and statistical analysis. Chapter 4 proposes practical recommendations for improving information system integration maturity in PMO environments based on the empirical findings.

The methodological basis of the study includes systematic literature review, survey-based quantitative research, structured expert interviews, descriptive statistical analysis, and correlation and regression analysis.

The study is expected to reveal that most PMO environments in Latvia operate at intermediate integration maturity levels, and that higher integration maturity is associated with greater reporting automation, better data quality, and stronger governance standardization. The findings are intended to provide evidence-based guidance for organizations seeking to improve their project management information systems.

The Bachelor Paper is written in **XX** pages and contains **YY** tables, **ZZ** figures, **xy** algorithms, **yz** listings, **xz** annexes, and a list of references with **zx** sources.

Keywords: Project Management Office, Information System Integration, Integration Maturity Model, Reporting Automation, Data Quality, Business Intelligence, Governance Standardization.

Key words / Atslēgvārdi

English

Project Management Office

Information System Integration

Integration Maturity Model

Reporting Automation

Data Quality

Business Intelligence

Governance Standardization

Latvian

Projektu vadības birojs

Informācijas sistēmu integrācija

Integrācijas brieduma modelis

Pārskatu automatizācija

Datu kvalitāte

Biznesa inteliģence

Pārvaldības standartizācija

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Introduction

Topicality of the problem: In the context of growing digital transformation, Project Management Offices (PMOs) in Latvian enterprises increasingly rely on multiple information systems for planning, reporting, and governance. However, many organizations struggle with fragmented system landscapes where data must be manually consolidated across tools, leading to inefficiencies, data quality issues, and delayed decision-making. Despite the practical significance of this challenge, empirical research on information system integration maturity in PMO environments within Latvia remains limited.

Aim of the study: The aim of the study is to assess the integration maturity of information systems within PMO environments in medium and large Latvian enterprises and examine its relationship with reporting automation, data quality, and governance standardization.

Object of the study: Information systems within Project Management Office environments in Latvian enterprises.

Subject of the study: The integration maturity of information systems and its relationship with reporting automation, data accuracy, and governance standardization in PMO environments.

Research problem: Many PMOs in Latvian enterprises operate with low levels of information system integration, resulting in poor architectural coherence, manual data consolidation, reduced reporting efficiency, and inconsistent governance practices.

Tasks of the study:

1. Conduct a systematic literature review on PMO digitalization, information system integration architectures, and maturity models relevant to IS environments.
2. Adapt and operationalize an Integration Maturity Model (Levels 1–4) with clearly defined, survey-measurable criteria for each level, based on integration mechanisms, automation level, reporting structure, and BI centralization.
3. Design and validate a structured survey instrument containing Likert-scale and categorical questions aligned with the Integration Maturity Model dimensions.

4. Collect empirical data through distribution of the survey to representatives of medium and large Latvian enterprises (50+ employees) with formal PMO structures, targeting 30–50 responses.
5. Conduct 2–3 structured expert interviews to supplement survey findings with architectural depth and qualitative insights.
6. Perform descriptive statistical analysis to determine the distribution of integration maturity levels across the sampled PMO environments.
7. Apply correlation analysis (Pearson or Spearman, depending on data distribution) and simple regression analysis to examine relationships between integration maturity and reporting automation, data accuracy, and governance standardization.
8. Interpret findings, draw conclusions regarding the hypotheses, and formulate practical recommendations for PMO information system integration improvement in Latvian enterprises.

The correspondence between the research tasks and the thesis chapters is presented in Table 1.

Table 1. Correspondence of research tasks to thesis chapters

Research Task	Thesis Chapter
Task 1 – Literature review on PMO digitalization and maturity models	Chapter 1
Task 2 – Design of Integration Maturity Model (Levels 1–4)	Chapter 1
Task 3 – Survey instrument design and validation	Chapter 2
Task 4 – Empirical data collection from Latvian enterprises	Chapter 3
Task 5 – Expert interviews	Chapter 3
Task 6 – Descriptive statistical analysis	Chapter 3
Task 7 – Correlation and regression analysis	Chapter 3
Task 8 – Conclusions and practical recommendations	Chapter 4

Hypotheses:

1. Most PMO environments in medium and large Latvian enterprises operate at intermediate (Levels 2–3) rather than advanced integration maturity levels.

2. PMOs utilizing API-based or platform-level integration demonstrate higher reporting automation than those relying primarily on manual data consolidation.
3. Integration maturity positively correlates with perceived data accuracy and timeliness.
4. The use of centralized BI/dashboard systems is associated with higher governance standardization in PMO processes.

Research methods: The study applies systematic literature review, survey-based quantitative research, structured expert interviews, descriptive statistical analysis, and correlation and regression analysis (Pearson or Spearman depending on data distribution).

Approbation of the study: To be completed upon presentation of research findings at a student conference or expert review.

1. Literature Review and Theoretical Background

1.1. Core Concepts and Definitions

This section defines the foundational terminology used throughout the thesis. Each term is defined once and used consistently in subsequent chapters.

1.1.1. Project Management Office (PMO)

A Project Management Office (PMO) is a centralized organizational unit responsible for defining and maintaining project management standards, governance frameworks, and reporting structures across an enterprise [11]. According to the Project Management Institute, PMOs can be classified into three types based on their degree of control and influence:

- **Supportive PMO** — provides templates, best practices, and guidance with minimal direct control over projects;
- **Controlling PMO** — enforces compliance with project management frameworks and requires the use of specific methodologies;
- **Directive PMO** — takes direct control of projects, with project managers reporting directly to the PMO.

In the context of this thesis, the term PMO refers to any formal organizational unit in medium or large Latvian enterprises that fulfills at least a controlling or directive function in project governance and reporting.

1.1.2. Project Management Information System (PMIS)

A Project Management Information System (PMIS) is a software-based system or combination of tools used to support the planning, execution, monitoring, and reporting of projects within a PMO environment [7]. Common PMIS tools include Microsoft Project, Jira, SAP, and Power BI. A PMIS may consist of a single integrated platform or multiple specialized tools operating in parallel, which introduces the challenge of information system integration.

1.1.3. Information System Integration

Information system integration refers to the process of linking two or more software systems so that data flows automatically between them without manual intervention [6]. In PMO environments, integration typically occurs between project tracking tools, financial systems, and reporting platforms. The degree of integration can range from fully manual data consolidation to fully automated, real-time data synchronization via Application Programming Interfaces (APIs) or unified platform ecosystems.

1.1.4. Integration Maturity Model

A maturity model is a structured framework that describes the progressive development of a capability across a defined set of levels [10]. In the context of information system integration, a maturity model defines the stages through which an organization advances from basic, manual integration practices toward fully automated, platform-level integration. This thesis adapts the concept of maturity modeling to the specific context of PMO information systems, proposing a four-level Integration Maturity Model as the primary theoretical framework.

1.1.5. Business Intelligence and Dashboard Systems

Business Intelligence (BI) refers to the technologies, processes, and practices used to collect, integrate, analyze, and present business data to support decision-making [9]. In PMO environments, BI systems typically manifest as centralized dashboards that aggregate data from multiple project and financial systems into a unified view. The presence or absence of centralized BI systems is used in this thesis as an indicator of governance standardization maturity.

1.1.6. Data Quality

Data quality refers to the degree to which data is accurate, complete, consistent, and timely for its intended purpose [2]. In the context of PMO reporting, data quality is directly influenced by the level of system integration: manually consolidated data is more prone to errors and delays, while automated integration reduces the risk of inconsistencies. This thesis measures data quality through perceived data accuracy and timeliness as reported by PMO representatives.

1.1.7. Reporting Automation

Reporting automation refers to the use of software tools and system integrations to generate project status reports, dashboards, and portfolio summaries without manual data entry or manipulation [11]. Automated reporting reduces the administrative burden on PMO staff and improves the timeliness and consistency of management information. In this thesis, reporting automation is operationalized as the extent to which PMO reports are generated automatically from integrated systems rather than compiled manually.

1.1.8. Medium and Large Enterprises in the Latvian Context

In accordance with the European Commission enterprise size classification, a medium enterprise is defined as an organization employing between 50 and 249 persons, while a large enterprise employs 250 or more persons [1]. This thesis targets medium and large Latvian enterprises as they are more likely to have established formal PMO structures and sufficiently complex information system landscapes to make integration maturity a relevant concern.

1.2. Related Work: Themes, Methods, and Evidence

This section surveys existing research on PMO digitalization, information system integration, and maturity models. Prior work is organized into five thematic strands, which are contrasted to identify the gap that the present thesis addresses.

1.2.1. Traditional PMO Research and Manual Integration Practices

Early research on Project Management Offices focused primarily on process standardization, governance frameworks, and the role of PMOs in improving project delivery outcomes [5]. In this body of literature, information systems were treated as secondary tools — typically spreadsheets and standalone reporting software — rather than integrated platforms. Manual data consolidation was the dominant practice, with PMO staff collecting data from multiple sources and compiling reports manually. While this approach established the theoretical foundations of PMO governance, it did not address the challenges of system fragmentation or the consequences of low integration maturity on reporting efficiency and data quality. The primary limitation of this strand is its failure to account for the growing complexity of modern PMO information system landscapes.

1.2.2. API-Based and Platform-Level Integration Research

A second strand of research examines the technical dimensions of information system integration, focusing on API-based connectivity and platform-level ecosystems [6]. Studies in this area demonstrate that organizations adopting API-driven integration architectures achieve significantly higher levels of reporting automation and data consistency compared to those relying on manual or file-based data transfer [7]. Platform ecosystems such as Microsoft 365, SAP, and Jira provide native integration capabilities that reduce the need for manual intervention. However, this research strand is predominantly technical in orientation and does not systematically examine the organizational or governance implications of integration maturity in PMO-specific contexts. Furthermore, implementation costs and the technical expertise required for API-based integration present significant barriers for medium-sized enterprises.

1.2.3. Business Intelligence and Governance Standardization

A third research strand investigates the relationship between Business Intelligence (BI) systems and organizational governance [9]. Studies consistently show that organizations with centralized BI dashboards demonstrate higher levels of governance standardization, as real-time data visibility enables more consistent decision-making and performance monitoring across projects [11]. This strand is particularly relevant to the present thesis, as it provides theoretical support for Hypothesis 4 — that the use of centralized BI systems is associated with higher governance standardization in PMO processes. The main limitation of this research strand is its focus on large enterprises, with limited attention to medium-sized organizations or specific regional contexts.

1.2.4. Maturity Model Frameworks in Information Systems Research

The fourth strand encompasses research on maturity models applied to information systems and organizational capabilities [10]. Established frameworks such as the Capability Maturity Model Integration (CMMI), the Organizational Project Management Maturity Model (OPM3), and the Portfolio, Programme and Project Management Maturity Model (P3M3) provide structured approaches to assessing and improving organizational capabilities across defined levels. More recent research has extended maturity modeling to digital transformation contexts, proposing Digital Maturity Models (DMMs) that assess the degree of digitalization across multiple organizational dimensions [8]. However, none of these frame-

works specifically address information system integration maturity in PMO environments as a distinct construct, leaving a gap that the present thesis seeks to fill.

1.2.5. Digital Maturity Research in Baltic and Eastern European Enterprises

The fifth strand encompasses emerging research on digital transformation and maturity in Baltic and Eastern European enterprises. Studies indicate that Latvian and broader Baltic enterprises generally demonstrate lower levels of digital maturity compared to Nordic counterparts, particularly in the areas of process automation and system integration [3]. While some research has examined e-government digitalization and SME digital adoption in Latvia, empirical studies specifically addressing information system integration maturity in PMO environments remain absent from the literature. This represents a significant gap, as the Latvian enterprise context — characterized by a predominance of medium-sized organizations and a rapidly evolving digital landscape — may exhibit distinct integration maturity patterns that differ from those documented in Western European or North American studies. The present thesis directly addresses this gap by providing the first empirical assessment of PMO IS integration maturity in Latvian enterprises.

Table 1.1. Comparison of related research strands

Strand	Focus	Strength	Limitation
Traditional PMO	Process standardization	Strong governance theory	No IS integration focus
API Integration	Technical connectivity	Automation evidence	No PMO context
BI & Governance	Dashboard systems	Governance link	Large enterprises only
Maturity Models	Capability levels	Structured framework	Not PMO IS specific
Baltic Digital Maturity	Regional context	Latvia relevance	No PMO IS empirics

1.3. Theoretical Framework

This section presents the theoretical lens that grounds the empirical investigation. Based on the review of existing maturity models and IS integration research in Section 1.2, this thesis proposes a four-level Integration Maturity Model (IMM) specifically adapted to PMO information system environments.

1.3.1. Foundations of the Integration Maturity Model

The proposed model draws on three theoretical foundations. First, the staged progression concept from the Capability Maturity Model Integration (CMMI) [10], which posits that organizations advance through discrete, measurable levels of capability. Second, the enterprise integration patterns framework [6], which classifies integration mechanisms by their degree of automation and architectural sophistication. Third, the data quality dimensions framework [2], which links integration architecture to data accuracy, completeness, and timeliness outcomes.

1.3.2. The Four-Level Integration Maturity Model

The Integration Maturity Model proposed in this thesis defines four levels of IS integration maturity in PMO environments, as presented in Table 1.2.

Table 1.2. Four-Level Integration Maturity Model for PMO Information Systems

Level	Name	Integration Mechanism	Reporting	Governance
1	Initial	No integration; fully manual data consolidation via spreadsheets	Fully manual; high error rate	Ad hoc; no standardization
2	Developing	File-based integration (CSV/Excel exports); partial automation	Semi-manual; scheduled exports	Basic standards; inconsistent
3	Defined	API-based point-to-point integration; automated data flows	Mostly automated; near real-time	Defined processes; mostly consistent
4	Optimized	Platform-level integration; unified data ecosystem with BI	Fully automated; real-time dashboards	Fully standardized; continuous improvement

Level 1 — Initial. At this level, PMO environments rely entirely on manual data consolidation. Project data is collected from multiple disconnected tools and compiled into reports manually, typically using spreadsheets. There is no systematic data flow between systems, resulting in high risk of errors, inconsistencies, and reporting delays. Governance practices are ad hoc and unstandardized.

Level 2 — Developing. At Level 2, some degree of semi-automated integration exists through file-based data exchange, such as scheduled CSV or Excel exports between systems. While this reduces some manual effort, significant human intervention is still required to

consolidate and validate data. Reporting is semi-automated and governance standards are partially defined but inconsistently applied.

Level 3 — Defined. Level 3 organizations have implemented API-based point-to-point integrations between key PMO systems, enabling automated data flows between project tracking, financial, and reporting tools. Reporting is mostly automated and near real-time. Governance processes are clearly defined and consistently followed across most PMO functions.

Level 4 — Optimized. At the highest level of maturity, PMO environments operate within a unified platform ecosystem where all systems are fully integrated and data flows in real time. Centralized BI dashboards provide complete portfolio visibility, and governance processes are fully standardized with continuous improvement mechanisms in place. This level represents the benchmark for information system integration excellence in PMO environments.

1.3.3. Variables and Relationships

The model defines the following key variables and relationships that will be tested empirically in Chapter 3:

- **Independent variable:** Integration Maturity Level (1–4), operationalized through survey items measuring integration mechanisms, automation level, and BI centralization.
- **Dependent variables:** Reporting automation degree, perceived data accuracy and timeliness, and governance standardization level.
- **Expected relationships:** Higher integration maturity levels are expected to be positively associated with all three dependent variables, consistent with the hypotheses stated in the Introduction.

1.4. Research Motivation and Direction

Problem significance. The fragmentation of information systems in PMO environments represents a significant operational challenge for Latvian enterprises. When project tracking, financial, and reporting tools operate in isolation, PMO staff are forced to spend substantial time on manual data consolidation, increasing the risk of errors and reducing the timeliness of management reporting. This inefficiency directly affects the quality of strategic decision-making at the portfolio level. The stakeholders most affected include PMO managers, project

sponsors, and executive leadership who rely on accurate and timely project data to make investment and resource allocation decisions.

Identified gap. As established in Section 1.2, existing research on IS integration maturity does not specifically address PMO environments, and no empirical study has examined this phenomenon in the Latvian enterprise context. The gap can be stated precisely as follows: *the distribution of IS integration maturity levels across PMO environments in Latvian enterprises, and its relationship with reporting automation, data quality, and governance standardization, has not been empirically measured or documented.* This gap is both testable and measurable through survey-based quantitative research.

Direction and scope. This thesis addresses the identified gap by conducting an empirical survey-based study of IS integration maturity in PMO environments across medium and large Latvian enterprises. The study applies the four-level Integration Maturity Model proposed in Section 1.3 as the primary measurement framework. The scope is deliberately bounded to enterprises with 50 or more employees that maintain a formal PMO structure, as smaller organizations are unlikely to have sufficiently complex IS landscapes to make integration maturity a meaningful concern. The study does not examine individual project outcomes, software implementation processes, or the technical architecture of specific integration solutions — these are explicitly out of scope. This direction directly corresponds to the Aim and Tasks defined in the Introduction.

Success criteria. The study will be considered successful if it achieves the following outcomes: (1) a valid and reliable survey instrument is developed and administered to at least 30 respondents from Latvian enterprises with formal PMO structures; (2) the distribution of integration maturity levels across the sample is described using descriptive statistics; (3) statistically significant relationships between integration maturity and the dependent variables (reporting automation, data quality, governance standardization) are tested using correlation and regression analysis; and (4) practical recommendations for PMO IS integration improvement are formulated based on the empirical findings.

1.5. Chapter Summary

This chapter established the theoretical and conceptual foundations of the thesis. Section 1.1 defined eight key terms — PMO, PMIS, IS integration, integration maturity model, business intelligence, data quality, reporting automation, and medium/large enterprise — providing a consistent conceptual vocabulary for the remainder of the thesis. Section 1.2 surveyed five thematic strands of related research, contrasting traditional PMO research, API integration studies, BI-governance research, maturity model frameworks, and Baltic digital maturity research, and identified a concrete empirical gap. Section 1.3 proposed a four-level Integration Maturity Model as the primary theoretical framework, defining the variables and relationships to be tested empirically. Section 1.4 articulated the problem significance, the measurable research gap, the scope of the study, and the success criteria.

Conclusions for Chapter 1. The literature review confirms that while substantial research exists on PMO governance, IS integration architectures, and digital maturity models, no empirical study has specifically examined IS integration maturity in PMO environments within Latvia. The proposed four-level Integration Maturity Model provides a theoretically grounded and operationalizable framework for addressing this gap. These foundations motivate the research methodology presented in Chapter 2 and the empirical study conducted in Chapter 3.

2. Research Methodology

This chapter presents the research methodology applied in this thesis. It describes the overall research design, the operationalization of the Integration Maturity Model, the design of the survey instrument and expert interviews, the data collection procedure, and the statistical methods used to analyze the collected data.

2.1. Research Design and Approach

This thesis adopts a quantitative research design, supplemented by qualitative expert interviews. The primary research strategy is a cross-sectional survey study, in which data is collected from multiple organizations at a single point in time. This design is appropriate for the research aim, which is to assess the current distribution of IS integration maturity levels across PMO environments in Latvian enterprises and examine its relationships with reporting automation, data quality, and governance standardization [12].

The philosophical stance of the study is post-positivist, assuming that an objective reality exists and can be measured through structured instruments, while acknowledging that measurement is subject to some degree of error. This stance is consistent with the use of Likert-scale survey items and statistical analysis as the primary data collection and analysis methods.

The research follows a deductive approach, in which the four hypotheses stated in the Introduction are derived from the theoretical framework presented in Chapter 1 and tested against empirical data collected from Latvian enterprises.

2.2. Operationalization of the Integration Maturity Model

The four-level Integration Maturity Model (IMM) proposed in Chapter 1 serves as the primary measurement framework. To enable survey-based measurement, each maturity level is operationalized through a set of observable, measurable indicators, as presented in Table 2.1.

Respondents are asked to select the description that best matches their organization's current PMO IS integration approach. The selected level serves as the primary independent variable in the analysis.

Table 2.1. Operationalization of Integration Maturity Model levels

Level	Name	Observable Indicators	Survey Items
1	Initial	No system connections; all data consolidated manually in spreadsheets; no automated reports	"Data is consolidated manually from multiple tools into spreadsheets"
2	Developing	Some file-based exports between systems; partial automation; scheduled report generation	"Some systems share data via file exports (CSV/Excel)"
3	Defined	API connections between key systems; automated data flows; near real-time reporting	"Systems are connected via APIs; reports are generated automatically"
4	Optimized	Full platform integration; unified BI dashboard; real-time portfolio visibility	"All systems are fully integrated in a unified platform with real-time dashboards"

2.3. Survey Instrument Design

The survey instrument was designed to measure the key constructs of the study: integration maturity level, reporting automation, data quality, and governance standardization. The instrument consists of four parts:

- **Part A — Organizational Profile:** Questions about the organization's size, industry, and PMO structure (categorical items).
- **Part B — Integration Maturity Assessment:** A single categorical item in which respondents select the IMM level that best describes their PMO's IS integration approach, supplemented by Likert-scale items measuring specific integration practices.
- **Part C — Reporting Automation and Data Quality:** Five-point Likert-scale items measuring the degree of reporting automation and perceived data accuracy and timeliness.
- **Part D — Governance Standardization:** Five-point Likert-scale items measuring the degree of governance standardization in PMO processes.

All Likert-scale items use a five-point response scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree). The survey was designed in English and distributed electronically via Google Forms. The estimated completion time is 10–15 minutes.

2.4. Survey Questionnaire

The complete survey instrument is presented in Table 2.2. Each item is aligned with a specific research construct and hypothesis.

Note on scaling. All items in Parts B–D use a five-point Likert scale: 1 = Strongly Disagree, 2 = Disagree, 3 = Neutral, 4 = Agree, 5 = Strongly Agree. Item C5 is reverse scored, meaning higher values indicate lower data quality issues.

2.5. Expert Interview Design

To supplement the quantitative survey findings with qualitative depth, 2–3 structured expert interviews are conducted with PMO leads, IT architects, or digital transformation managers from Latvian enterprises. The interviews follow a structured protocol with open-ended questions covering the following themes:

- The current state of IS integration in the respondent’s PMO environment
- Challenges experienced due to low integration maturity
- The role of BI and dashboard systems in governance standardization
- Perceived barriers to advancing integration maturity
- Practical recommendations for improving PMO IS integration

Interview responses are analyzed qualitatively using thematic analysis to identify patterns and insights that complement the quantitative findings.

2.6. Data Collection Procedure

The survey is distributed to representatives of medium and large Latvian enterprises (50+ employees) with formal PMO structures. The target sample size is 30–50 responses, which is sufficient for the planned descriptive and correlational statistical analyses [4]. Respondents are identified through professional networks, LinkedIn, and direct organizational contacts. Only one response per organization is accepted to ensure independence of observations.

Data collection takes place over a period of four to six weeks. All responses are anonymous and stored securely in accordance with data protection principles. Participation

is voluntary and respondents are informed of the purpose of the study prior to completing the survey.

2.7. Data Analysis Methods

The collected data is analyzed using the following statistical methods:

Descriptive statistics. Frequency distributions and percentages are used to describe the distribution of integration maturity levels across the sample. Mean scores and standard deviations are calculated for Likert-scale items measuring reporting automation, data quality, and governance standardization.

Normality testing. The Shapiro-Wilk test is applied to determine whether the data follows a normal distribution, which informs the choice between parametric (Pearson) and non-parametric (Spearman) correlation analysis [4].

Correlation analysis. Pearson or Spearman correlation coefficients are calculated to examine the relationships between integration maturity level and the three dependent variables: reporting automation, data quality, and governance standardization. Statistical significance is assessed at the $p < 0.05$ level.

Regression analysis. Simple linear regression is applied to examine the predictive relationship between integration maturity level and each dependent variable, allowing for quantification of the strength and direction of the relationships.

Qualitative analysis. Expert interview transcripts are analyzed using thematic analysis to identify recurring themes and insights that supplement the quantitative findings.

All quantitative analyses are conducted using SPSS or R statistical software.

2.8. Validity and Reliability

To ensure the quality of the research instrument and findings, the following measures are taken to address validity and reliability.

Content validity. The survey items were designed based on established theoretical frameworks — specifically the Integration Maturity Model proposed in Chapter 1, the CMMI framework [10], and the data quality dimensions from DAMA-DMBOK [2]. Each survey item is explicitly linked to a research construct and hypothesis, ensuring that the instrument measures what it is intended to measure.

Construct validity. The survey constructs — integration maturity, reporting automation, data quality, and governance standardization — are operationalized based on definitions

from peer-reviewed literature and established frameworks. Multiple items per construct are included to reduce measurement error and improve construct coverage.

Internal consistency reliability. Cronbach's alpha coefficient will be calculated for each multi-item construct scale to assess internal consistency. A Cronbach's alpha value of 0.70 or above is considered acceptable for research purposes [4].

External validity. The study targets a specific population — representatives of medium and large Latvian enterprises with formal PMO structures. While the findings may not be directly generalizable to all organizations globally, they are intended to be representative of the Latvian enterprise context within the defined scope.

2.9. Research Ethics

This study adheres to the following ethical principles throughout the data collection and analysis process:

- **Voluntary participation.** All survey respondents and interview participants participate voluntarily and may withdraw at any time without consequence.
- **Informed consent.** Respondents are informed of the purpose of the study, how their data will be used, and who will have access to the results before completing the survey.
- **Anonymity and confidentiality.** Survey responses are collected anonymously. No personally identifiable information is collected or stored. Interview participants are anonymized in the thesis unless explicit consent for identification is provided.
- **Data protection.** All collected data is stored securely and processed in accordance with the General Data Protection Regulation (GDPR) and applicable Latvian data protection legislation.
- **Honest reporting.** All findings are reported accurately and without manipulation, including results that do not support the research hypotheses.

2.10. Limitations of the Research Design

While this study is designed to provide a rigorous empirical assessment of IS integration maturity in PMO environments in Latvia, several limitations must be acknowledged.

Self-reported data. The survey relies on self-reported perceptions of PMO representatives, which may be subject to social desirability bias — respondents may rate their

organization's integration maturity higher than it actually is. To mitigate this, the survey includes specific behavioral indicators rather than general self-assessment questions.

Sample size. The target sample of 30–50 responses is sufficient for descriptive and correlational analysis but limits the statistical power of regression models and prevents subgroup analysis by industry or organization size. Future research with larger samples would strengthen the generalizability of findings.

Cross-sectional design. The study captures a snapshot of integration maturity at a single point in time and cannot establish causal relationships between integration maturity and the dependent variables. Longitudinal research would be needed to confirm causal direction.

Geographic scope. The study is deliberately bounded to Latvian enterprises, which limits the generalizability of findings to other national or regional contexts. However, this scope is appropriate given the research aim of providing Latvia-specific empirical evidence.

Construct measurement. Integration maturity is measured through a self-developed instrument rather than a previously validated scale, as no existing validated scale specifically measures IS integration maturity in PMO environments. While the instrument is grounded in established theoretical frameworks, its psychometric properties require further validation in future research.

2.11. Chapter Summary

This chapter described the research methodology applied in this thesis. A quantitative cross-sectional survey design with qualitative expert interview supplementation was adopted. The four-level Integration Maturity Model was operationalized into measurable survey indicators. The survey instrument was designed to measure integration maturity, reporting automation, data quality, and governance standardization across PMO environments in Latvian enterprises. Data collection targets 30–50 responses from medium and large Latvian enterprises with formal PMO structures. Statistical analysis methods including descriptive statistics, normality testing, correlation analysis, and regression analysis are applied to test the four research hypotheses. Validity, reliability, ethical principles, and research limitations were also addressed. The results of the empirical study are presented in Chapter 3.

Table 2.2. Survey instrument items by construct and hypothesis

No.	Survey Item	Construct	Hypothesis
Part A — Organizational Profile			
A1	How many employees does your organization have? (50–249 / 250+)	Organization size	–
A2	What industry does your organization operate in? (categorical)	Industry sector	–
A3	How long has your PMO been formally established? (years)	PMO maturity	–
A4	How many active projects does your PMO currently oversee? (numerical)	PMO scope	–
Part B — Integration Maturity Assessment			
B1	Which of the following best describes how data flows between your PMO’s information systems? (select one IMM level description)	Integration maturity level	H1
B2	Our PMO systems share data automatically without manual intervention.	Integration automation	H1, H2
B3	Our PMO uses API connections between project tracking, financial, and reporting tools.	API integration	H2
B4	Our PMO relies on manual data entry or file exports (CSV/Excel) to consolidate project data.	Manual integration	H1, H2
B5	Our PMO operates within a unified platform ecosystem where all systems are fully integrated.	Platform integration	H1, H2
Part C — Reporting Automation and Data Quality			
C1	Project status reports in our PMO are generated automatically from integrated systems.	Reporting automation	H2
C2	Our PMO dashboards are updated in real time without manual intervention.	Reporting automation	H2
C3	The data in our PMO reports is accurate and reliable.	Data accuracy	H3
C4	Project data is available in a timely manner for decision-making.	Data timeliness	H3
C5	We frequently encounter errors or inconsistencies in our PMO reporting data. (reverse scored)	Data quality	H3
Part D — Governance Standardization			
D1	Our PMO follows standardized processes for project reporting across all projects.	Governance standardization	H4
D2	Our PMO uses centralized dashboards to monitor all projects in one place.	BI centralization	H4
D3	Project governance processes in our PMO are consistently applied across	Governance consistency	H4

3. Empirical Study / Practical Research (Example Chapter 3)

This chapter usually presents the applied or empirical part of the thesis: an experiment, prototype, survey, simulation, or case study. However, the exact scope depends on your project. In some theses, Chapters 2 and 3 are merged into one; in others, this chapter may be split into two separate ones (e.g., “Implementation” and “Evaluation”).

3.1. Design of the Study (Example Section)

Explain how your empirical/practical work was organized. Typical points to include (adapt as needed):

- **Objectives.** What you wanted to achieve or test.
- **Setup.** Hardware, software, datasets, or infrastructure used.
- **Participants or subjects (if any).** For surveys, experiments, or usability tests.
- **Procedures.** How the study/experiment was carried out, in enough detail to be reproducible.

Note. If your work is implementation-heavy, this section can describe the system architecture, algorithms, or workflows you built. If it is more research-focused, describe the methodology and design choices.

3.2. Results of the Study (Example Section)

Present the main outcomes clearly and systematically. Examples:

- Tables or charts with performance benchmarks, error rates, or runtime.
- Screenshots or diagrams of a prototype or interface.
- Summaries of survey/interview data.

Tip. Do not over-interpret here — keep deep discussion for Chapter 4. The focus here is on presenting what you observed.



Source: Replace with your own chart or prototype screenshot.

Figure 3.1. Example chart of study results (dummy figure).

3.3. Evaluation and Interpretation (Optional)

In some topics it is useful to add a subsection where you briefly evaluate your results against expectations or benchmarks. Examples: compare against a baseline algorithm, test statistical significance, or highlight unexpected patterns. If this does not fit your project, you can skip or merge it into the next chapter.

3.4. Sub-Conclusions (for Chapter 3)

End with a short summary (2–3 paragraphs). Example points:

- Key empirical findings (what was achieved, measured, or demonstrated).
- Relation to the aim and hypothesis from the Introduction.
- How these findings prepare for the next step — either a discussion chapter (Chapter 4) or the final conclusions (Chapter 6).

Reminder. The conclusions here should be short and local to this chapter. The overall “big picture” conclusions must still be saved for the dedicated **Conclusions chapter**.

4. Discussion and Development (Example Chapter 4)

This chapter may serve several possible purposes depending on the nature of your work: (1) discuss your empirical findings in more depth, (2) compare results against theory or prior work, (3) propose improvements, models, or design solutions, or (4) evaluate feasibility or efficiency. If your project is small-scale or self-contained, you may skip this chapter and move directly to the Conclusions.

4.1. Extended Discussion (Optional Example Section)

Here you may reflect more deeply on your results than in Chapter 3. Possible angles include:

- Comparing results with other studies or benchmarks.
- Highlighting unexpected patterns or anomalies.
- Theoretical implications: how your findings support or contradict existing models.

Tip. If your work is primarily an implementation, you may instead use this section to evaluate design trade-offs, architectural choices, or lessons learned.

4.2. Proposals / Improvements (Optional Example Section)

In many IT theses, the contribution is not only empirical but also practical. Here you can suggest:

- Proposed system modifications or design improvements.
- Prototype features, extensions, or optimizations.
- Conceptual models or frameworks based on your findings.

Illustrate with diagrams, mockups, or tables if useful. Remember: these are suggestions, not required elements.

4.3. Evaluation of Feasibility / Efficiency (Optional Example Section)

Sometimes it makes sense to estimate how well your proposal would work in practice.

Examples:

- Performance evaluation (e.g., runtime, memory use, scalability).
- Cost/benefit trade-offs (time, resources, usability).
- Risks and limitations of implementation.

This section can be skipped if it does not apply to your project.

4.4. Sub-Conclusions (for Chapter 4)

Close with a short synthesis that ties together the discussion and proposals. Example points:

- What your deeper reflection adds beyond the raw results.
- Which improvements or models are most promising.
- How these insights prepare the way for the **Conclusions chapter**.

Reminder. The “big picture” statements — confirmation of tasks, validation of the hypothesis, and achievement of the aim — belong in the **Conclusions chapter**, not here.

5. Discussion, Limitations, and Future Work (Optional Example Chapter 5)

This chapter is meant as an *extra space* for broader reflection. It can be useful if you want to go beyond the mandatory Conclusions and show awareness of the bigger picture. However, it should remain concise and focused — do not just repeat results already covered in previous chapters.

5.1. Extended Discussion (Optional)

Here you may reflect on your findings in greater depth. Examples of what to include:

- Compare your results to other studies or real-world applications.
- Highlight unexpected outcomes or open questions.
- Discuss theoretical or practical implications that extend beyond your specific project.

Tip. This section can be especially valuable if your work connects to ongoing research, industrial practice, or societal issues.

5.2. Limitations of the Study (Optional)

Every project has constraints — being open about them increases credibility. Possible categories:

- **Scope.** Narrow focus, specific use case, or limited generalizability.
- **Data/technical.** Small datasets, noisy logs, hardware/software constraints.
- **Methodological.** Simplifying assumptions, lack of longitudinal data, limited metrics.

Note. Do not exaggerate — just acknowledge realistically what was and wasn't possible within your timeframe and resources.

5.3. Future Work and Recommendations (Optional)

Suggest what could be done after your thesis. Examples:

- Follow-up studies or new research questions raised by your findings.
- Practical improvements to your system, prototype, or methods.
- Opportunities for scaling up, applying to other domains, or testing with different datasets.

This section is not required, but it can demonstrate initiative and forward thinking.

Final Note

Remember: the **mandatory Conclusions chapter** must already confirm the research tasks, validate the hypothesis, and state whether the aim of the thesis has been achieved. This optional chapter is only for broader reflection and forward-looking remarks — use it only if it adds real value.

6. Conclusions

The **Conclusions** chapter must clearly demonstrate that the aim and tasks of the Bachelor Paper have been fulfilled. This chapter is not a repetition of earlier text, but a synthesized summary that links the research aim, tasks, hypothesis, and results.

When preparing this section, students should ensure that:

- **Confirmation of tasks.** Explicitly state that all tasks set out in the Introduction have been completed. A good practice is to connect each task with the chapter where it was accomplished. Example: “Task 1 (literature review) was completed in Chapter 1, providing a theoretical framework for the study. Task 2 (analysis of the case/system) was carried out in Chapter 2. Task 3 (empirical study) was presented in Chapter 3. Task 4 (development of proposals) was fulfilled in Chapter 4.”
- **Validation of hypothesis.** Clearly indicate whether the research hypothesis (if formulated) was validated or rejected, based on the obtained results. Example: “The proposed hypothesis was confirmed, as the empirical results demonstrated ...”
- **Achievement of the aim.** Explicitly state that the aim of the thesis has been reached, supported by the fact that all tasks were accomplished and the hypothesis was addressed. Example: “Therefore, it can be concluded that the aim of the thesis has been achieved.”
- **Avoid repetition.** Do not copy entire findings from chapters. Instead, summarize them concisely to show the logical progression from tasks → results → aim.
- **Practical and theoretical value.** Conclude with a short statement on the significance of the results (e.g., how the findings may be used in practice or contribute to the field).

Recommended final formulation: “All tasks set out in the Introduction were accomplished (Tasks 1–4 correspond to Chapters 1–4). The research hypothesis was validated, and the aim of the thesis has been reached.”

Literature

- [1] “Commission recommendation of 6 may 2003 concerning the definition of micro, small and medium-sized enterprises,” European Commission, Accessed: Mar. 22, 2026. [Online]. Available: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32003H0361>
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A. LaTeX Hints

This appendix shows small, copy-pasteable examples that compile with this thesis class out of the box. It avoids any template-specific commands from other classes.

A.1. Paragraphs and inline emphasis

Write one sentence per source line (helps with version control). A blank line starts a new paragraph.

You can write *emphasized (italics)* and **bold** text. Use `\url{...}` for clickable links, e.g. `\url{https://ctan.org}`.

A.2. Math and equations

Inline math: $f(x) = x^2 + 1$. Numbered display equations use `amsmath`:

$$\int_0^1 x^2 dx = \frac{1}{3}. \tag{A.1}$$

We can reference equation (A.1) thanks to `cleveref`.

A.3. Figures and subfigures

A normal figure:



Figure A.1. Example figure.

Two images side-by-side via `subcaption`:

We can reference figures A.1, A.2 and A.2a.

A.4. Tables

A compact table with `booktabs`:

A long table across multiple pages (`longtable`):

(a) Left

(b) Right

Figure A.2. Two subfigures.

Table A.1. Compact example table

Factor	Mean	SD
Pay	2.0	0.5
Recognition	4.0	0.7
Growth	5.0	0.6

Table A.2. Long table example

Item	Group	Value
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1

Item	Group	Value
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3
A	Alpha	1
B	Alpha	2
C	Beta	3

A landscape figure page (pdfscape):



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Figure A.3. Landscape example.

A.5. Algorithms and code listings

Pseudocode with algorithm+algpseudocode:

Algorithm 1 Greedy selection (example)

```
1: Initialize  $S \leftarrow \emptyset$ 
2: while feasible choice exists do
3:   choose best feasible item  $x$ 
4:    $S \leftarrow S \cup \{x\}$ 
5: end while
6: return  $S$ 
```

Code with listings:

Listing A.1. Bubble sort

```
1 def bubble(a):
2     n = len(a)
3     for i in range(n-1):
4         for j in range(n-1-i):
5             if a[j] > a[j+1]:
6                 a[j], a[j+1] = a[j+1], a[j]
```

We can reference algorithm 1 and listing A.1.

A.6. TikZ: Trie data structure diagram

TikZ is a powerwfull diagraming, graphing engine allowing you to script creation of the diagrams. Refer to <https://tikz.dev/> for online documentation and examples.

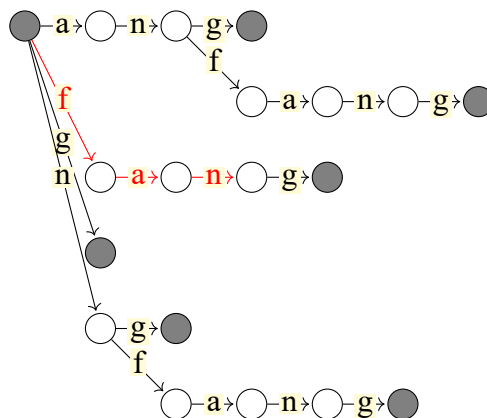


Figure A.4. Trie data structure demonstrating graph-based TikZ syntax with highlighted path.

If pgfplots is installed (we auto-load it if present), a quick plot:

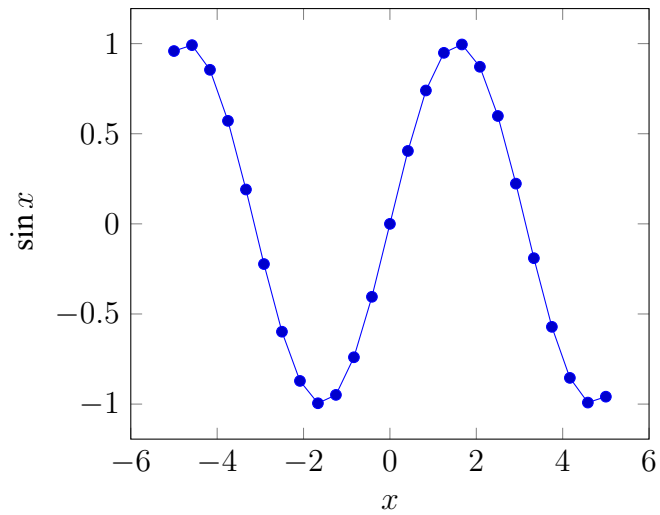


Figure A.5. Sine plot with pgfplots.

A.7. Cross-references

Use `\label{...}` and `\cref{...}` for equations, figures, tables, algorithms, and listings to get correct names and numbers automatically.

A.8. Bibliography citation

Cite like **porter2008** and list all sources in the “Literature” section (managed by biblatex with the IEEE style in this project).

B. Expert Evaluation Criteria for Final Thesis Quality

Table B.1. Expert evaluation criteria for the quality of the final thesis

No.	Criterion title and content	Points
1	Conformity of the thesis title to the research area; style; correctness of key words.	5
2	Thesis structure: presence of the essential chapters.	5
3	Level of research motivation grounded in the contemporary state of the problem.	5
4	Quality and correctness of the theses to be defended; novelty of the main results.	5
5	Adequacy of the research methodology to the formulated research problem; appropriateness of the theoretical framework.	5
6	Originality of the research problem.	5
7	Use of constructive methods (including mathematical and computational) in solving the research problems.	5
8	Quality of conclusions: justification and correctness.	5
9	Quality of thesis formatting, literary style, compliance with standards and regulations.	5
10	Quality of the list of references: currency, authority, presence of sources in foreign languages.	5

Note.

1. All criteria are mandatory.
2. Final thesis evaluation:
 - < 25 points — the thesis does not meet the requirements;
 - 25–35 points — the thesis requires improvement;
 - 36–50 points — the thesis is eligible for defense.

Apliecinājums / Affirmation

Ar šo es, Amina Showkath, apliecinu, ka bakalaura darbs ir izpildīts patstāvīgi, bez citu palīdzības, no svešiem avotiem ņemtie dati un definējumi ir uzrādīti darbā. Šis darbs nekādā veidā nav iesniegts nevienai citai pārbaudījuma komisijai un nekur nav publicēts.

(Hereby I, Amina Showkath, affirm that the Bachelor's thesis was performed independently; sources of data and definitions are provided. This work has not been submitted to any other examination commission and has not been published elsewhere.)

Rīga, 20____. _____.

paraksts/signature, atšifrējums/decription

Pateicības / Acknowledgments

Autors izsaka pateicību darba vadītājam par sniegtajiem padomiem un atbalstu, kā arī uzņēmuma “X” kolēģiem par palīdzību datu iegūšanā un aptaujas realizēšanā.

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